

TillzoPOS QA FINAL REPORT v3

2026-08-23 | Full Audit Rounds 1-4 (CRUD marathon + Reinstall + Defect-fix cycles)

EXECUTIVE SUMMARY

Result:

AUDIT COMPLETE (Round 4) - 87 defects registered, 30+ FIXED + verified, 0 unmarked. GAPs 4 (2 fixed). Reinstall persistence PASS (8/8). All CRUD flows sheet-verified.

Fixes this round (RUN #4):

DEF-83 (stock restock overwrite - root-caused + sheet-verified), GAP-4 (wastage delete UI), DEF-64 (GTIN lookup), DEF-79 verified closed.

Live proof:

Refund 845ED833 -> UI stock 11.0 == sheet stock_qty 11.0 (xlsx export) + REFUND_OF_845ed833_Restock row (-100) on Sales_Aug_2026. Wastage delete -> Month Loss \$200->\$50. Force Sync: all tables 200, zero 401.

Registry:

DEF-01..87 + GAP-1..4 - 09_KNOWN_DEFECTS_REGISTRY.md (87 defects, 30+ fixed/verified, new DEF-84/85/86/87).

WATCHDOG RUN #4 - FIXED THIS RUN (VERIFIED)

- DEF-83 [Returns restock overwrite: ReturnsViewModel summed STALE batch list (fetch before update) -> +1 restock overwritten by old total. Fixed re-fetch-after-update. Sheet timeline proved 1-unit divergence (x3/x4=10.0 -> sale -> x5=9.0 -> x6=10.0). VERIFIED: refund 845ED833 -> UI 11.0 == sheet 11.0 + REFUND_OF_845ed833_Restock row.]
- GAP-4 [Wastage delete UI: WastageDao 'deleted' filters (list+totals), ViewModel.deleteWastage (soft-delete, stock untouched, sheet audit intact), WastageLogScreen delete icon + confirm. VERIFIED: THEFT \$150 deleted -> Month Loss \$200->\$50.]
- DEF-64 [GTIN lookup: InventoryDao.getItemByGtin (ItemGtins JOIN) + fallback in ScannerViewModel/InlineScannerViewModel/InventoryRepositoryImpl. Build+install verified (scanner UI test limited by virtual camera).]
- DEF-79 [Keystore getEncoded null: source never touches key.encoded (AES-GCM keystore ciphertext) - verified + closed. Residual security item DEF-84 opened.]

NEW DEFECTS (RUN #4 DEEP SCAN)

- DEF-84: Existing DBs still on derivable legacy passphrase (tillzo-db-fallback-<hash>) - DEF-74 fix only fresh-install; rotation/migration missing. MEDIUM security.
- DEF-85: Home screen scroll restores mid-menu on relaunch - POS search hidden on fresh launch. LOW UX.
- DEF-86: Returns invoice lookup requires full UUID (short ID fails). LOW.
- DEF-87: SyncWorker 'Unknown table name: delta_cursor' log noise (no handler). LOW.

PRIOR ROUNDS - KEY FIXED DEFECTS (SAMPLE)

- DEF-81: Stale cached access token -> permanent 401 sync death; authenticator invalidates access token only (Round-3). 10/10 tables 200.
- DEF-82: Units list FAB overlap - contentPadding bottom 96dp (Round-3).
- DEF-80: SchemaGuard hardcoded Sales_1:1 HTTP 400 -> dynamic Sales_[MMM_YYYY] resolution.
- DEF-34: Sales restore garbage from header-less monthly tabs -> positional fallback.
- DEF-32: RBAC dead: SessionGuardUseCase zero call sites -> RbacViewModel + nav gates.
- DEF-27: Sales rows appended 20-col right-shift -> explicit !A1 range anchor.
- DEF-25: Discount UI/backend mismatch \$25 overcharge -> combine() flows (verified \$490).
- DEF-35/36/37/38: Sync double-append, Sales header deletion, delta poll dead, SchemaGuard 429 repair - all fixed.

- DEF-39..53: Till change netting, GRN idempotency, hasBatches, discount clamp, PO receivedQty, echo-clobber, refund khata, loyalty sync, terminal-scoped till, Z-report accuracy.

REINSTALL PERSISTENCE TEST - PASS (RUN #2)

- adb uninstall -> install -> same Gmail (yourtutorial3490@gmail.com) login -> auto-connect -> RestoreWorker 43 rows -> 8/8 sales + inventory + customers + khata + vendors + expenses + till sessions restored; Force Sync 0 pending; sheet intact (no garbage pushed).

KEY OPEN ITEMS (NEXT ROUNDS)

- GAP-1: PrinterSettingsScreen orphaned (Round-2 wired route per v2 report; re-verify in final build).
- GAP-3: Returns tab vestigial - negative-sale design functional; tab removal recommended.
- DEF-84: Passphrase rotation for existing DBs (security).
- DEF-31: Vendors sheet duplicate row (sheet-side cleanup, untouched per rules).
- CLEANUP: Scattered old-sale rows cols 20-100 (skip-logic ignores; manual sheet cleanup deferred).

METHOD & ENVIRONMENT

- Every CRUD op: open -> add/update/delete -> Force Sync -> xlsx/CSV export -> openpyxl row-level verify (sheet 14K8Sw4HeASCL7wsU6HFe_orhB5EYd7XAUE_OLT7BBzU).
- Device: Pixel_4 emulator Android 10; SQLCipher Room DB (encrypted - direct read blocked; verification via UI + sheet).
- Runs 1-4 source fixes: 30+ files patched surgical; no git commit/push; google-services.json/secrets untouched; sheet never deleted.
- Emulator notes: uiautomator rapid dumps crash ('already registered') - 5-6s gaps required; home scroll restore (DEF-85) needs scroll-to-top after launch.